Simultaneous Equations and Inequalities

ANSWER KEY

Section 1: Linear-Quadratic Simultaneous Systems

1 Solve the simultaneous system $y - 2x = 3$ and $x^2 + y^2 = 26$ for the positive solution pair (x, y).

a. $(1, 5)$ ✓

b. $(2, 7)$

c. $(3, 9)$

d. $(4, 11)$

Substitute $y = 2x + 3$:
$x^2 + (2x + 3)^2 = 26 \Rightarrow 5x^2 + 12x - 17 = 0 \Rightarrow (5x + 17)(x - 1) = 0 \Rightarrow x = 1, y = 5$.

2 Find the two points of intersection of the line $2x + y = 7$ and the rectangular hyperbola $xy = 6$.

a. $(2, 3)$ **and** $\left(\frac{3}{2}, 4\right)$ ✓

b. $(1, 5)$ and $(3, 1)$

c. $(2, 3)$ and $(3, 2)$

d. $(4, -1)$ and $(1, 6)$

$x(7 - 2x) = 6 \Rightarrow 2x^2 - 7x + 6 = 0 \Rightarrow (2x - 3)(x - 2) = 0 \Rightarrow (2, 3)$ and $(3/2, 4)$.

3 Determine the value of k for which the line $x + y = k$ is tangent to the curve $y = \frac{4}{x}$.

a. $k = \pm 4$ ✓

b. $k = 4$ only

c. $k = \pm 2$

d. $k = 8$

$x(k - x) = 4 \Rightarrow x^2 - kx + 4 = 0$. $\Delta = k^2 - 16 = 0 \Rightarrow k = \pm 4$.

Section 2: Quadratic Inequalities in One Variable

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Find the set of values of x satisfying the quadratic inequality $2x^2 - 5x - 3 \geq 0$.

a. $x \leq -\frac{1}{2}$ **or** $x \geq 3$ ✓

b. $-\frac{1}{2} \leq x \leq 3$

c. $x < -1$ or $x > 3$

d. $x \geq 3$ only

$(2x + 1)(x - 3) \geq 0$. Parabola opens upward: $x \leq -1/2$ or $x \geq 3$.

5 Solve the strict quadratic inequality $x(x + 4) < 12$.

a. $-6 < x < 2$ ✓

b. $x < -6$ or $x > 2$

c. $-2 < x < 6$

d. $0 < x < 3$

$x^2 + 4x - 12 < 0 \Rightarrow (x + 6)(x - 2) < 0 \Rightarrow -6 < x < 2$.

6 Find the range of values of x for which the expression $9 - x^2$ is strictly positive.

a. $-3 < x < 3$ ✓

b. $x < -3$ or $x > 3$

c. $x < 3$

d. $0 < x < 9$

$9 - x^2 > 0 \Leftrightarrow x^2 < 9 \Leftrightarrow -3 < x < 3$.

Section 3: Rational Inequalities and Sign Analysis

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 Solve the rational inequality $\frac{x-2}{x+3} \leq 0$.

a. $-3 < x \leq 2$ ✓

b. $-3 \leq x \leq 2$

c. $x < -3$ or $x \geq 2$

d. $x \leq 2$

Sign changes at $x = -3$ and $x = 2$. Denominator cannot be 0, so $x \in (-3, 2]$.

8 Find the complete solution set for the inequality $\frac{2x+1}{x-1} > 1$.

a. $x < -2$ **or** $x > 1$ ✓

b. $-2 < x < 1$

c. $x > 1$ only

d. $x > -2$

$\frac{2x+1-(x-1)}{x-1} = \frac{x+2}{x-1} > 0 \Rightarrow x < -2$ or $x > 1$.

9 Solve the inequality $\frac{1}{x} < 2$ for all non-zero real x.

a. $x < 0$ **or** $x > \frac{1}{2}$ ✓

b. $0 < x < \frac{1}{2}$

c. $x > \frac{1}{2}$ only

d. $x < \frac{1}{2}$

Multiply by $x^2 > 0$: $x < 2x^2 \Rightarrow x(2x-1) > 0 \Rightarrow x < 0$ or $x > 1/2$.

Section 4: Systems with Modulus and Geometric Applications

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 Find the values of x satisfying the simultaneous conditions $|x-3| \le 4$ and $x^2 - 4x > 0$.

a. $-1 \le x < 0$ **or** $4 < x \le 7$ ✓

b. $-1 \le x \le 7$

c. $0 < x < 4$

d. $4 < x \le 7$ only

$|x-3| \le 4 \Rightarrow [-1, 7]$. $x(x-4) > 0 \Rightarrow x < 0$ or $x > 4$. Intersection: $[-1, 0) \cup (4, 7]$.

11 The perimeter of a right-angled triangle is 30 cm and its hypotenuse is 13 cm. Find the lengths of the two legs.

a. 5 cm **and** 12 cm ✓

b. 6 cm and 11 cm

c. 7 cm and 10 cm

d. 8 cm and 9 cm

$a + b = 17$, $a^2 + b^2 = 169 \Rightarrow 2ab = 17^2 - 169 = 120 \Rightarrow ab = 60$. Legs are 5 and 12.

12 Find the range of values of x where the line $y = 3x - 2$ lies entirely above the parabola $y = x^2 - 2x + 4$.

a. $2 < x < 3$ ✓

b. $x < 2$ or $x > 3$

c. $-3 < x < -2$

d. No such interval exists

$3x - 2 > x^2 - 2x + 4 \Rightarrow x^2 - 5x + 6 < 0 \Rightarrow (x-2)(x-3) < 0 \Rightarrow 2 < x < 3$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647